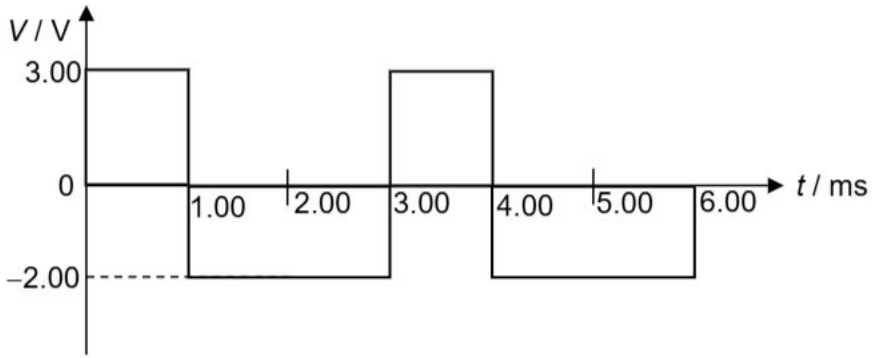
24 The graph below shows how the voltage of an a.c. source V varies with time t.

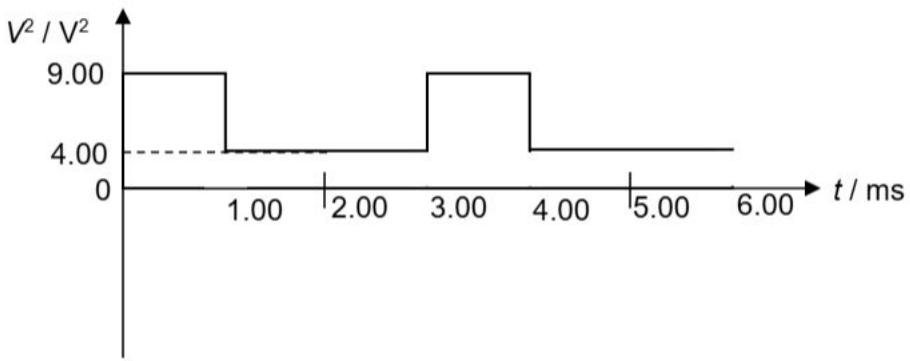


What is the r.m.s. voltage of the source?

A 2.12 V **B** 2.38 V **C** 2.50 V **D** 2.55 V

Ans: B

First square the V-t graph.



Next find the mean square voltage.

$$\langle V^2 \rangle = \frac{9.00 + (2 \times 4.00)}{3}$$
$$= 5.6667$$

Lastly, find the r.m.s. voltage.

$$\text{r.m.s. } V = \sqrt{5.6667}$$
$$= 2.38 \text{ V}$$